

SS-RAIM based Integrity Architecture for CDGNSS Systems against Satellite Measurement Faults

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Abstract

In this paper, we propose an integrity architecture for the Carrier Phase-based Differential Global Navigation Satellite System (CDGNSS). The CDGNSS utilizes carrier-phase measurements and is reliant on the correct estimation (or resolution) of cycle ambiguities. However, due to the nonlinear nature of ambiguity resolution, protection level computation for CDGNSS presents greater challenges compared to that of traditional code-based systems. The method we propose addresses these difficulties by employing Solution Separation-based Receiver Autonomous Integrity Monitoring (SS-RAIM). This architecture considers the integrity risk associated with the incorrect ambiguity resolution by including correct/incorrect ambiguity fix probabilities into the protection level equation. In order to apply the SS-RAIM to CDGNSS, the main contributions of this work, which have not been explored in the previous works, are twofold. First, we revisit the variance computation property of the solution separations-based detection test statistic. Second, we verify the applicability of the measurement overbounding method. These considerations, which arise from the differences in navigation algorithms between code-based and CDGNSS systems, are mathematically demonstrated. Finally, we present performance analysis results using an illustrative example of a dual-constellation, dual-frequency navigation system.

1. INTRODUCTION

Carrier Phase-based Differential Global Navigation Satellite Systems (CDGNSS) are important for applications requiring centimeter levels of accuracy, such as autonomous driving cars, drones and near shore maritime applications. CDGNSS uses carrier-phase measurements and relies on accurate estimation (or resolution) of cycle ambiguities. Ambiguity resolution is the process of resolving the unknown cycle ambiguities of the carrier phase measurements as integers. Once this is done successfully, very precise carrier-phase measurements act as very precise range measurement as well. However, incorrectly resolved integer ambiguities may result in unacceptably large position errors. Therefore, the CDGNSS integrity architecture must limit and mitigate the risk of incorrect ambiguity resolution to enable its use in safety-critical applications.

Safety-assured CDGNSS systems must be designed to comply with (sub-allocated) integrity requirements under the fault free hypothesis (H0), ground receiver failure hypothesis (H1), and satellite measurement failure hypothesis (H2). Pervan and Chan (2001) developed a method to classify incorrect integer ambiguity resolution within the H0 hypothesis and to account for integrity risk associated with it by defining a required probability of correct ambiguity resolution. This method conservatively assumed that any incorrect ambiguity resolution causes loss of integrity, meaning that the position error exceeds a predefined alert limit for a specific navigation operation without warning. Since this assumption limits the availability of the system, Khanafseh and Pervan (2008) and Wu et al. (2008) developed several methods to loosen this conservatism by evaluating the integrity risk caused by incorrect integers directly in the position domain. Green and Humphreys (2019) proposed a data-driven method to reduce the other conservatism arising from the need to protect against potential incorrect fixes without the benefit of conditioning on the observed measurements for H0 hypothesis. In addition, Khanafseh et al. (2013) proposed a Solution Separation-based Receiver Autonomous Integrity Monitoring (SS-RAIM)-based method to protect against receiver faults for the H1 hypothesis. However, the aforementioned previous works do not address the H2 failure which is one of the main contributors of integrity loss.

The H2 hypothesis includes atmospheric propagation anomalies (e.g., large ionospheric and tropospheric errors) and signal-in-space faults (e.g., erroneous ephemeris data). One viable approach to ensuring H2 integrity of CDGNSS is to use augmentation-system monitoring separate from users in a manner similar to Ground-based Augmentation Systems (GBAS) and Satellite-based Augmentation Systems (SBAS). An example of a GBAS-based integrity-guaranteed CDGNSS system is Shipboard Relative GPS (SRGPS), which was developed to support automatic shipboard landings in zero-visibility conditions (Rife et al., 2008). In it, the base station detects faults using its own monitors, and the corrections broadcast to users indicate which ranging measurements (if any) are unsafe to use and thus need to be excluded. Because the datalink between base station and users in GBAS and SBAS-related approaches is one-way only (from base station to users), existing external monitors rely solely on the information available at the base station to detect satellite faults. Differential errors in the range measurements caused by certain undetected faults increase as the baseline distance between the user and base station increases. These errors might be tolerable for certain applications such as the SRGPS application, for which the baseline distance is short. However, they might not be acceptable for all CDGNSS applications, such as a lane-keeping system of autonomous cars or delivery drones, whose baseline distances could be several tens of kilometers or more. This dependency of bias on the baseline length significantly limits the coverage of integrity-guaranteed CDGNSS systems.

Another feasible approach for H2 faults is to include monitoring at each user, such as RAIM. In simple terms, RAIM compares one satellite measurement to the consensus of the other available satellite measurements. Since the onboard monitor is contained within the user receiver, it incorporates measurements from both the user and the base station. As a result, the detectability of monitors increases and the remaining biases do not depend on the baseline length. El-Mowafy and Kubo (2017), El-Mowafy and Kubo (2018), and Wang and El-Mowafy (2021) have each applied RAIM to CDGNSS. These prior works used a solution separation method to detect faults and derive protection level equations. However, they did not consider the impact of incorrect ambiguity resolution when computing protection levels. They instead assumed that incorrect integer ambiguity resolution is impossible once internal ambiguity validation tests (e.g., ratio tests) are passed. This assumption is not strictly true because the ratio test only assesses the closeness of a float solution to its nearest integer, not the correctness of the integer ambiguity (Teunissen & Verhagen, 2009). Therefore, the impact of incorrect ambiguity resolution on the position solution must be quantified to derive protection level equations, even if the ratio test passes.

In response, this paper proposes a SS-RAIM based integrity architecture for CDGNSS against H0 and H2 hypotheses. The proposed concept accounts for the impact of nominal measurement errors and satellite measurement failure on both position estimate error and ambiguity resolution. The proposed architecture, detailed in Section II, is based on the system developed by Khanafseh et al. (2013). The main contributions of this work, described in Section III and IV, include the derivation of the covariance matrix equation for the monitor detection statistic and the application of the overbounding method to upper-bound the integrity risk. Although these methods were originally developed for code-based systems, the differences in navigation algorithms between code-based and CDGNSS systems necessitate their reconsideration for CDGNSS. This area has not been explored in previous works regarding CDGNSS integrity.

Section III describes the statistical characteristics of solution separation-based detection statistic for CDGNSS system. In code-based SS-RAIM system, it has been derived that the variance matrix for the monitor detection statistics can be computed by subtracting the variance matrix of the all-in-view solution from that of the fault-free solution. This property facilitates the easy computation of the covariance matrix of the detection statistic, and thus significantly simplifies the evaluation of the continuity risk (Joerger et al., 2014). In CDGNSS system, this variance computation property, if it remains valid, could enable efficient computation of the continuity risk evaluation. However, the validity of this property is not trivial matter due to the differences on the navigation algorithm. The filter states of code-based systems consist of elements common to all satellites (e.g., position), whereas the filter states of CDGNSS include satellite-dependent elements (e.g., carrier-phase ambiguities) as well as common elements. Furthermore, the pivot satellite concept for measurement double-differencing, which is not utilized in code-based system, is used for CDGNSS filter. These differences in the navigation filter necessitate us to newly derive the variance computation property of SS-RAIM detection statistics. In this context, Section III directly derives the covariance matrices of SS-RAIM detection statistics as a function of covariance matrices of fixed position estimates for CDGNSS systems, and demonstrates the same property of the code-based systems holds valid for CDGNSS systems.

Section IV verifies the validity of the measurement model overbounding method for the CDGNSS. This method, originally developed for code-based systems, provides an upper bound on the integrity risk in cases where the probability distributions of measurement noise are unknown but can be upper bounded. Because the integrity risk of a code-based system is computed by a one-dimensional integral over a constant range of integration (i.e., where the magnitude of position error exceeds the protection level),

only the integrand is a concern. However, the evaluation of integrity risk for CDGNSS includes a multi-dimensional integral. In this context, Section IV demonstrates the applicability of these overbounding methods to CDGNSS. Finally, the proposed method is evaluated in Section V and conclusions are provided in Section VI.

2. SS-RAIM BASED INTEGRITY ASSURED ARCHITECTURE FOR CD-GNSS SYSTEM

The ambiguity resolution, key to achieving fast and high-precision CD-GNSS positioning, involves resolving the unknown cycle ambiguities of the carrier-phase measurement as integers. Once the ambiguity has been successfully resolved, the precise carrier-phase measurement acts as a very accurate range measurement, enabling precise navigation. However, incorrect resolution can lead to unacceptable positioning errors. Hence, for safety-critical applications, it becomes essential to develop a rigorous methodology for computing a statistical bound of the position error. This bound, referred to as a Protection Level (PL) in the aviation community, is derived from the associated integrity risk requirement for the specific navigation operation. It should be derived by quantifying the impact of the measurement errors on the ambiguity resolution.

The effect of the measurement errors on the ambiguity resolution process is very complex. In code-based systems, since the entirety of the navigation algorithm processes, such as least square filter or Kalman filter, is linear, the position estimate follows a normal distribution if the measurement errors are normally distributed. Thus, the dispersion of the position error can be entirely captured by the filter covariance matrix. This enables us to evaluate PL through a traditional straight forward covariance analysis. However, since the navigation processes of CDGNSS involve non-linear processes due to ambiguity resolution, the position error does not follow a normal distribution but rather a multimodal normal distribution (Teunissen, 2002). Therefore, traditional covariance analysis is not directly applicable for quantifying the impact of the measurement errors on position error following ambiguity resolution.

In this context, we propose the integrity assurance architecture of CDGNSS considering the non-linear nature of the ambiguity resolution. This architecture is based on the work of Kanafseh et al. (2013). Section 2.1 describes the outline of the architecture, and the integrity risk and continuity risks are discussed in Section 2.2 and 2.3, respectively

2.1. Architecture of CDGNSS System

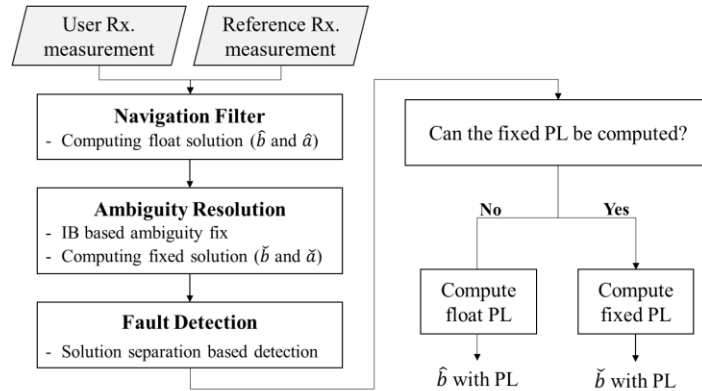


Figure 1. Schematic diagram of the proposed integrity architecture of CDGNSS system

Figure 1 presents a schematic diagram of our proposed method to guarantee the integrity against the H0 and H2 hypotheses in CDGNSS system. The raw measurements of the user and reference receivers are fed into the navigation filter (e.g., least-square filter, or Kalman filter). These inputs are processed to compute the float baseline vector estimate ($\hat{b}_0$) and float double differenced ambiguity estimate ($\hat{a}_0$) between the user and the reference station, and along with their variance matrices, $Q_{\hat{b}_0}$ and $Q_{\hat{a}_0}$, respectively. These estimates are the solutions obtained by discarding the integer nature of the ambiguities (Teunissen & Montenbruck, 2017). The integer constraints are exploited during the ambiguity resolution step to obtain a better estimate of the baseline. This step involves mapping from the space of real numbers to the space of integer numbers, converting $\hat{a}_0$ to corresponding integer values ($\check{a}_0$). Several integer mapping methods exist, but this work uses integer bootstrapping (IB). IB is a popular method which combines conditional least-squares estimation with rounding to sequentially estimate the components of the integer vector (Teunissen, 2002). It is straightforward to implement and has a closed-form expression for the probability of successful integer resolution. Once $\check{a}_0$ is

obtained, the ambiguity residual (i.e., $\hat{a}_0 - \check{a}_0$) is used to readjust $\hat{b}_0$ to obtain the so-called fixed baseline ($\check{b}_0$) solution. The quality of $\check{b}_0$ aligns with the high precision of the carrier phase data, provided the probability of $\check{a}_0$ being the correct integer is sufficiently high (Teunissen & Montenbruck, 2017).

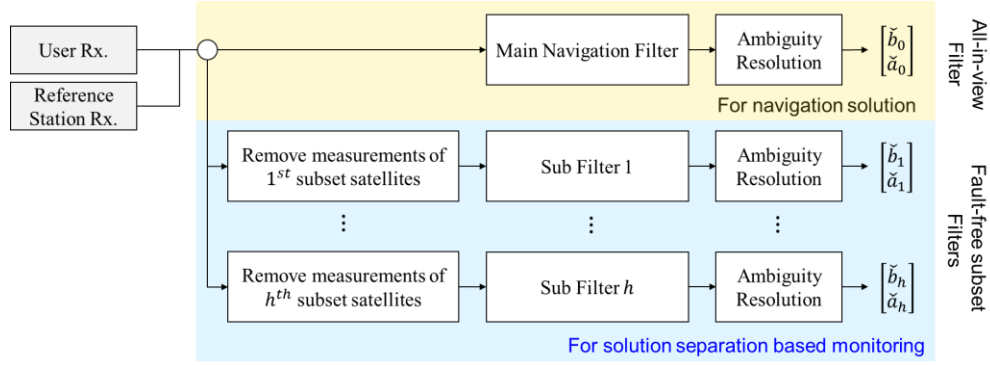


Figure 2. Parallel filters for SS-RAIM

To guarantee the integrity of $\check{b}_0$ against the H2 failure modes, a fault list to be monitored should be established. This list consists of H_i hypothesis where the i^{th} subset of all-in-view satellites is faulted for $i \in \{0, \dots, h\}$, and is determined based on the priori probability of satellite fault (Juan et al., 2015). The subscript 0 is used to designate the fault free hypothesis (H_0). We use the solution separation-based detection statistics (Δ_i) to detect the $H_{i \neq 0}$ fault, which are defined as:

$$\Delta_i = \check{b}_i - \check{b}_0 \quad (1)$$

The subscript i identifies the fault hypothesis of H_i and designates its corresponding fault-free solution ($\check{b}_i$). The $\check{b}_i$ is obtained from a sub-filter which does not use the measurements of the i^{th} subset satellites, as shown in Figure 2. The subscript 0 indicates the fault-free hypothesis (H_0), and it is also used to designate the all-in-view navigation solution ($\check{b}_0$). The $\check{b}_0$ is computed under all hypotheses, but is not fault-free under H_i for $i \neq 0$. The monitor issues an alert when any Δ_i exceeds its corresponding predefined threshold (T_i). The T_i is determined from the false alert risk, detailed in Section 2.3.

When no faults are detected, the PL of $\check{b}_0$ is computed to ensure the integrity against the undetected faults. The impact of undetected fault on the ambiguity resolution should be taken into account during PL computation since it can lead an incorrect ambiguity resolution. However, quantifying this impact, without the specific knowledge of the fault characteristics (e.g., its magnitude, direction), is complex due to the non-linearity of ambiguity resolution process. Therefore, we derive a conservative PL of $\check{b}_0$ by using $\check{b}_i$ which remains unaffected by the fault under H_i hypothesis and T_i which is used to detect the fault, in order to avoid the computation of the impact of the faults on $\check{b}_0$. Further details are provided in Section 2.2.

Another factor that can lead to incorrect ambiguity resolution is the nominal measurement noise, such as multipath error. The impact of this noise on the ambiguity resolution decreases as the filtering time increases. However, before the filter has sufficiently converged, a frequent incorrect ambiguity resolution can occur due to the measurement noise, which results in position biases. These biases may or may not lead to integrity and continuity losses depending on the size of ambiguity error. We conservatively assume that all incorrect ambiguity resolutions, regardless of their size, result in continuity and integrity losses (Kanafesh et al., 2013). The purpose of this assumption is not only to reduce the computation load, but also to provide an upper bound on the integrity risk without the perfect statistical knowledge of measurement error, which will be detailed in Section 4.

This conservative assumption enhances computational efficiency and enable conservative integrity assessment. However, this could potentially limit the availability of the CDGNSS system. When the float ambiguity is not accurate enough, the probability of incorrect ambiguity resolution might exceed the integrity and continuity risk requirements. This results in the system failing to meet the integrity and continuity risk requirements because all incorrect ambiguity resolutions are assumed to lead to integrity and continuity losses. Therefore, in this case, the float baseline solution $\hat{b}_0$, which is not affected by ambiguity resolution, should be

used as an alternative as shown in Figure 1. Since there are many researches on the integrity and continuity of $\hat{b}_0$, such as (Joerger et al, 2014), we focus on $\check{b}_0$ in this work.

2.2. Integrity Risk

Integrity is related to the trust that can be placed in the information provided by the navigation system. The integrity risk is defined as the probability that the position error exceeds an error bound without a notice from the fault monitor (Pullen, 2011). The error bound which satisfies the integrity risk requirement is referred to PL. In the proposed architecture, the monitor does not issue an alert only when all Δ_i do not exceed the threshold. Thus, the vertical PL (VPL) is defined as (Joerger et al., 2014):

$$\sum_{i=0}^h P\left(|\check{b}_{0,v} - b_v| > VPL, \bigcap_{j=1}^h |\Delta_{j,v}| < T_{j,v} \mid H_i\right) \cdot P_{H_i} = I_{req,v} \quad (2)$$

where the subscript v represents the vertical component and $I_{req,v}$ is the integrity risk requirement allocated to vertical coordinate. The b is a true baseline vector, and the P_{H_i} is a priori probability of H_i . In this paper, we focus on the vertical coordinate, but it can be easily extended to the horizontal coordinate.

Evaluating Equation (2) necessitates calculating the integrity risk under not only fault-free condition but also fault conditions. Computing the integrity risk under fault condition presents a challenge due to insufficient information about the fault. The given condition H_i provides information about which satellites are at fault, but does not specify its magnitude and direction. This limitation makes it complicate to determine the distribution of $\check{b}_{0,v}$ and $\Delta_{j,v}$. Thus, we utilize the following upper bound:

$$P\left(|\check{b}_{0,v} - b_v| > VPL, \bigcap_{j=1}^h |\Delta_{j,v}| < T_{j,v} \mid H_i\right) < P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL \mid H_i) \quad (3)$$

Since $\check{b}_i$ is a fault free solution under H_i hypothesis, the upper bound probability can be computed without the knowledge of the fault. This upper bound was originally derived for the code-based SS-RAIM system (Joerger et al., 2014), but it was established without any assumptions on the distribution of navigation solutions and detection statistics. Therefore, this bound is valid for $\check{b}_i$ and Δ_i for $i \in \{0, \dots, h\}$, even if they do not follow the same distribution as those in the code-based system.

For code-based systems, position estimates typically follow a normal distribution when measurement noises are normally distributed. However, this does not hold for CDGNSS systems due to the non-linearity nature of the ambiguity resolution process (Teunissen, 2002). Many prior works usually rely on the assumption that the fixed ambiguity is deterministic, and consider that the fixed baseline solution follows a normal distribution. However, as highlighted by Teunissen (2002), this is not strictly correct. The fixed ambiguity solution is estimated from the measurements and since the measurements are modelled as random variates, it is random variates too. Therefore, we employ the multiple hypothesis approach, which was used in (Teunissen, 2002), to evaluate Equation (3):

$$P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL \mid H_i) = \sum_{z_i \in \mathbb{Z}^{n_{a_i}}} P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL \mid \check{a}_i = z_i, H_i) \cdot P(\check{a}_i = z_i \mid H_i) \quad (4)$$

where n_{a_i} is the number of elements of $\check{a}_i$. The case where $\check{a}_i$ is equal to a_i , which is a true ambiguity, corresponds to correct ambiguity fix, whereas the other cases correspond to incorrect ambiguity fix. The $P(\check{a}_i = z_i \mid H_i)$ is computed from the covariance matrix of $\hat{a}_i$ (Teunissen, 2006).

Under the given condition of $\check{a}_i = z_i$, the $\check{a}_i$ is a deterministic value, not a random variate. Thus, the conditional baseline estimate (i.e., $\check{b}_i \mid \check{a}_i = z_i, H_i$) follows, when the measurement errors are normally distributed, a normal distribution with a mean of $\mu_{\check{b}_i}(z_i)$ and covariance matrix of $Q_{\check{b}_i \mid \check{a}_i}$ (Teunissen, 2002):

$$\mu_{\check{b}_i \mid \check{a}_i}(z_i) = Q_{\hat{b}_i \hat{a}_i} Q_{\hat{a}_i}^{-1} (z_i - a_i) \quad (5)$$

$$Q_{\hat{b}_i|\check{a}_i} = Q_{\hat{b}_i} - Q_{\hat{b}_i\hat{a}_i}Q_{\hat{a}_i}^{-1}Q_{\hat{b}_i\hat{a}_i}^T$$

where $Q_{\hat{b}_i}$ and $Q_{\hat{a}_i}$ are the covariance matrices of $\hat{b}_i$ and $\hat{a}_i$, respectively. $Q_{\hat{b}_i\hat{a}_i}$ is a correlation matrix between $\hat{b}_i$ and $\hat{a}_i$. While the $Q_{\check{b}_i|\check{a}_i}$ does not depend on z_i , the $\mu_{\check{b}_i|\check{a}_i}(z_i)$ is a function of z_i . The conditional baseline estimate is unbiased only when $z_i = a_i$ (i.e., correct fix).

In Equation (4), because the set of all possible z_i constitutes a countably infinite set, the exact evaluation is impossible. This leads us to use the following upper bound which is obtained from the fact that $\sum_{z_i \in \mathbb{Z}} P(\check{a}_i = z_i) = 1$:

$$P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL | \check{a}_i = a_i, H_i) \cdot P(\check{a}_i = a_i | H_i) + (1 - P(\check{a}_i = a_i | H_i)) \quad (6)$$

This upper bound follows from a conservative assumption where all incorrect ambiguity resolution results in the integrity loss (i.e., $P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL | \check{a}_i = z_i, H_i) = 1$ for $z_i \neq a_i$) (Kanafesh et al., 2013). It is quite conservative because some $z_i (\neq a_i)$ may induce a small bias on $\check{b}_i$ without any severe impact on the integrity risk.

The purpose behind this conservative assumption is not just for computational efficiency, but more importantly, to establish an upper bound for the integrity risk, even in the absence of precise knowledge about the statistical characteristics of the measurement errors. The integrity risk in Equation (6) is computed from the error covariance matrices of the baseline and ambiguity. To ensure that the integrity risk calculation is reliable, these matrices must be correct. However, it is impossible since the covariance matrices derive from the statistical models of measurement errors which are never exactly known. Therefore, using the model without a thorough consideration may lead to optimistic assessments of integrity risk, potentially resulting in severe consequences in safety. Section 4 demonstrates that, even without perfect knowledge of the measurement model, the conservative assumption used for Equation (6), when combined with the measurement overbounding method, can provide an upper bound for the integrity risk.

The solution to the following equation, which is obtained by substituting the Equation (6) into Equation (2), provides a VPL that meets the required integrity risk:

$$\sum_{i=0}^h [P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL | \check{a}_i = a_i, H_i) \cdot P(\check{a}_i = a_i | H_i) + (1 - P(\check{a}_i = a_i | H_i))] \cdot P_{H_i} = I_{req,v} \quad (7)$$

There are several methods available to solve Equation (7). We used a half-interval search algorithm proposed by Juan et al. (2015).

In Equation (7), it is important to note that the following constraint must be satisfied in order to determine the VPL from Equation (7):

$$\sum_{i=0}^h (1 - P(\check{a}_i = a_i)) \cdot P_{H_i} < I_{req,v} \quad (8)$$

The term $1 - P(\check{a}_i = a_i)$ represents the probability of incorrect fix of $\check{a}_i$. This constraint imposes that the entire incorrect fix probability (which is the sum of contributions of each sub filter) stays below the integrity risk requirement. Since the incorrect fix probability decreases when the precision of $\hat{a}_i$ gets better, the constraint can be met as the filtering time increases.

2.3. Continuity Risk

Continuity risk is defined as the probability of unexpected interruption in navigation after initiation of an operation (Pullen, 2011). These interruptions can arise from events falling under two categories: detections under a fault-free hypothesis, and detections under fault hypotheses. The detection threshold, T_i , is generally set based on a continuity risk requirement allocated to the fault free hypothesis in order to limit the likelihood of false alarms (Joerger et al., 2014). In our proposed system, an alarm is triggered whenever

any Δ_i exceeds its corresponding T_i . Therefore, in the absence of the fault-exclusion algorithm, T_i can be defined from the following false alert risk equation (Joerger et al., 2014):

$$P\left(\bigcup_{i=1}^h |\Delta_{i,v}| > T_{i,v} | H_0\right) = C_{req,0,v} \quad (9)$$

where $C_{req,0}$ is the continuity risk requirement allocated to H_0 hypothesis, and $C_{req,0,v}$ is vertical portion of $C_{req,0}$. A direct evaluation of the false alert risk presents a challenge because the Δ_i s for $i \in \{1, \dots, h\}$ are correlated (Joerger et al., 2014). In practice, an upper-bound is used to determine T_i (Joerger et al., 2014):

$$\sum_{i=1}^h P(|\Delta_{i,v}| > T_{i,v} | H_0) \cdot P_{H_0} = C_{req,0,v} \quad (10)$$

This bound allows the evaluation of the false alert risk for each test statistic, individually.

In contrast to the code base system where Δ_i follows a zero-mean normal distribution under the H_0 hypothesis, the Δ_i of CDGNSS does not follow a normal distribution. This deviation is resulted from the fact that $\check{b}_i$ and $\check{b}_0$ do not follow normal distribution. Thus, to evaluate the false alert risk, multiple hypothesis approach is employed:

$$P(|\Delta_{i,v}| > T_{i,v} | H_0) = \sum_{z_i \in \mathbb{Z}^{n_{a_i}}} \sum_{z_0 \in \mathbb{Z}^{n_{a_0}}} P(|\Delta_{i,v}| > T_{i,v} | \check{a}_i = z_i, \check{a}_0 = z_0, H_0) \cdot P(\check{a}_i = z_i, \check{a}_0 = z_0 | H_0) \quad (11)$$

Since both conditional $\check{b}_i$ and $\check{b}_0$, given condition of $\check{a}_i = z_i$ and $\check{a}_0 = z_0$, follow the normal distribution, the conditional Δ_i (i.e., $\Delta_i | \check{a}_i = z_i, \check{a}_0 = z_0$) follows the normal distribution with the mean of $\mu_{\Delta_i | \check{a}_i, \check{a}_0}(z_i, z_0)$ and covariance matrix of $Q_{\Delta_i | \check{a}_i, \check{a}_0}$:

$$\begin{aligned} \mu_{\Delta_i | \check{a}_i, \check{a}_0}(z_i, z_0) &= Q_{\check{b}_i \check{a}_i} Q_{\check{a}_i}^{-1}(z_i - a_i) - Q_{\check{b}_0 \check{a}_0} Q_{\check{a}_0}^{-1}(z_0 - a_0) \\ Q_{\Delta_i | \check{a}_i, \check{a}_0} &= Q_{\check{b}_i | \check{a}_i} - Q_{\check{b}_0 | \check{a}_0} \end{aligned} \quad (12)$$

The $\mu_{\Delta_i | \check{a}_i, \check{a}_0}(z_i, z_0)$ derives from Equation (1) by taking the expectation operator, and is zero only when $z_i = a_i$ and $z_0 = a_0$. The derivation of $Q_{\Delta_i | \check{a}_i, \check{a}_0}$ will be detailed in Section 3.

The exact computation of Equation (11) is impossible due to the countably infinite set of z_i and z_0 . Hence, an upper bound is derived by using the fact that $\sum_{z_i \in \mathbb{Z}^{n_i}} \sum_{z_0 \in \mathbb{Z}^{n_0}} P(\check{a}_i = z_i, \check{a}_0 = z_0 | H_0) = 1$:

$$P(|\Delta_{i,v}| > T_{i,v} | \check{a}_i = a_i, \check{a}_0 = a_0, H_0) \cdot P(\check{a}_i = a_i, \check{a}_0 = a_0 | H_0) + (1 - P(\check{a}_i = a_i, \check{a}_0 = a_0 | H_0)) \quad (13)$$

This equation derives from the assumption which implies that any incorrect fix on $\check{a}_i$ or $\check{a}_0$ results in the continuity loss (i.e., $P(|\Delta_{i,v}| > T_{i,v} | \check{a}_i = z_i, \check{a}_0 = z_0, H_0) = 1$ for $z_i \neq a_i$ or $z_0 \neq a_0$) (Khanafseh et al., 2013).

From Equation (13), the $T_{i,v}$ which satisfies the continuity risk requirement allocated to H_0 is determined as:

$$T_{i,v} = \Phi^{-1}\left(\frac{C_{req,0,i,v}/P_{H_0} - (1 - P(\check{a}_i = a_i, \check{a}_0 = a_0 | H_0))}{P(\check{a}_i = a_i, \check{a}_0 = a_0 | H_0)}\right) \cdot \sigma_{\Delta_{i,v} | \check{a}_i, \check{a}_0} \quad (14)$$

where $\Phi^{-1}(\cdot)$ is the inverse tail probability distribution of the two-tailed standard normal distribution (i.e., $\Phi(\cdot) = 1 - \text{normcdf}(\cdot)$, where $\text{normcdf}(\cdot)$ is the standard normal cumulative distribution function). $\sigma_{\Delta_{i,v} | \check{a}_i, \check{a}_0}^2$ is a vertical component of $Q_{\Delta_i | \check{a}_i, \check{a}_0}$. $C_{req,0,i,v}$ is an allocated probability of $C_{req,0,v}$ for $T_{i,v}$. While any arbitrary allocation ensures that the continuity risk requirement is met, we

equally allocated $C_{req,0,v}$ to each $T_{i,v}$ (i.e., $C_{req,0,i,v} = C_{req,0,v}/h$). It is important to note from Equation (14) that $T_{i,v}$ can be determined only when the term of $1 - P(\check{\alpha}_i = a_i, \check{\alpha}_0 = a_0|H_0)$ is smaller than $C_{req,0,i,v}$.

In Equation (14), the $P(\check{\alpha}_i = a_i, \check{\alpha}_0 = a_0|H_0)$ should be computed from the joint distribution of $\hat{a}_i$ and $\hat{a}_0$ to consider their correlation. This calculation, however, might be cumbersome. Therefore, a lower bound for $P(\check{\alpha}_i = a_i, \check{\alpha}_0 = a_0|H_0)$ is used for the computation of this joint probability (Khanafseh et al., 2013):

$$P(\check{\alpha}_i = a_i, \check{\alpha}_0 = a_0|H_0) \geq P(\check{\alpha}_i = a_i|H_0) \cdot P(\check{\alpha}_0 = a_0|H_0) \quad (15)$$

The use of this lower bound results in an upper bound for T_i , which ensures the continuity risk requirement allocated to H_0 .

3. THE VARIANCE OF SOLUTION SEPARATION BASED DETECTION STATISTIC FOR CDGNSS

In order to determine the detection threshold from Equation (14), the conditional variance of Δ_i under the given condition that $\check{\alpha}_i = a_i$ and $\check{\alpha}_0 = a_0$, is needed. In code-based SS-RAIM, the variance of solution separation-based detection statistics was computed by subtracting the variance of all-in-view solution from that of fault-free solution. This section demonstrates the property, which is shown in Equation (12), holds valid for CDGNSS.

In Section 3.1, under the condition that $\check{\alpha}_i$ is given $\mathbf{z}_i$, we express the fixed baseline solutions and its corresponding covariance matrices as functions of the time sequence of measurements, satellite geometry matrices, and partitioning matrices. In Section 3.2, $Q_{\Delta_i|\check{\alpha}_i, \check{\alpha}_0}$ will be directly derived from the covariance matrices of fixed baseline solutions.

3.1. Fixed Baseline Solution Given $\check{\alpha}_i = \mathbf{z}_i$

A linear dynamic system is described at any discrete time k of a time-sequence, by a process equation and a measurement equation:

$$\begin{aligned} \text{Process Equation:} \quad & b_k = b_{k-1} + w_{b,k} \\ & a_{i,k} = a_{i,k-1} \end{aligned} \quad (16)$$

$$\begin{aligned} \text{Measurement Equation:} \quad & D_i^T N_i^T \rho_k = D_i^T N_i^T B_k b_k + D_i^T N_i^T \varepsilon_{\rho,k} \\ & D_i^T N_i^T \phi_k = D_i^T N_i^T B_k b_k + \lambda a_{i,k} + D_i^T N_i^T \varepsilon_{\phi,k} \end{aligned} \quad (17)$$

where the subscript i represents the fault hypothesis H_i . ρ and ϕ are $m \times 1$ dimensional vectors, representing the single differenced (between the receivers) code and carrier measurements, respectively. b is 3×1 dimensional baseline vector. a_i is $(m - n_i - 1) \times 1$ dimensional double-differenced ambiguity vector of i^{th} sub-filter, where n_i is the number of faulted satellites under H_i hypothesis. w_b is 3×1 dimensional vector assumed normally distributed with zero-mean and covariance matrix of W_b . In this work, W_b is assumed as a diagonal matrix whose diagonals are infinite since there is no information for the propagation from b_{k-1} to b_k . B is a satellite geometry matrix, and λ is a wavelength. ε_{ρ} and ε_{ϕ} are the $m \times 1$ dimensional single differenced code and carrier measurement error vectors, respectively. These error vectors are assumed to be normally distributed with zero-mean and constant covariance matrices of $\frac{1}{\alpha} I_m$ for code, and $\frac{1}{\beta} I_m$ for carrier measurements, where I_m is m dimensional identity matrix. D_i^T is a $(m - n_i - 1) \times (m - n_i)$ dimensional double differencing matrix utilized for the i^{th} sub-filter, defined as (Teunissen, 1997):

$$D_i^T = \begin{bmatrix} I_{p_i-1} & -1_{(p_i-1) \times 1} & 0_{(p_i-1) \times (m-n_i-p_i)} \\ 0_{(m-n_i-p_i) \times (p_i-1)} & -1_{(p_i-1) \times 1} & I_{m-n_i-p_i} \end{bmatrix} \quad (18)$$

when p_i -th satellite is used as a pivot satellite. $1_{n \times m}$ and $0_{n \times m}$ are $n \times m$ matrices, with all elements being ones and zeros, respectively. Note that the reason for the inclusion of D_i^T in Equation (17) lies in the fact that it directly shows the impact of pivot

satellite on the baseline solution (Teunissen, 1997). N_i^T is a partitioning matrix to isolate the fault free subset measurements from the entirety of measurements under the H_i hypothesis. Without loss of generality (given the order in which measurements are stacked in ρ and ϕ is arbitrary), it is assumed that, under the H_i hypothesis, the faulty measurements are the first n_i elements of ρ and ϕ . Thus, N_i^T can be expressed as (Joerger et al., 2014):

$$N_i^T = [0_{(m-n_i) \times n_i} \quad I_{m-n_i}] \quad (19)$$

Given a time sequence of measurements and state dynamics, the current time state estimates of the Kalman filter is identical to those of batch least square filter. The measurements stacked in a batch can be expressed in a single equation, which is much easier to analyze than a Kalman filter:

$$\begin{bmatrix} 0 \\ 0 \\ D_i^T N_i^T \rho_1 \\ D_i^T N_i^T \phi_1 \\ \vdots \\ 0 \\ 0 \\ D_i^T N_i^T \rho_k \\ D_i^T N_i^T \phi_k \end{bmatrix} = \begin{bmatrix} I & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ D_i^T N_i^T B_1 & 0 & \dots & 0 & 0 \\ D_i^T N_i^T B_1 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & -I & I \\ 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & D_i^T N_i^T B_k \\ 0 & 0 & \dots & 0 & D_i^T N_i^T B_k \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_{k-1} \\ b_k \end{bmatrix} + \begin{bmatrix} 0 & 0 & \dots & 0 & 0 \\ I & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ \lambda I & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & -I & I \\ 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & \lambda I \end{bmatrix} \begin{bmatrix} a_{i,1} \\ \vdots \\ a_{i,k-1} \\ a_{i,k} \end{bmatrix} + \begin{bmatrix} w_{b,1} \\ 0 \\ D_i^T N_i^T \varepsilon_{\rho,1} \\ D_i^T N_i^T \varepsilon_{\phi,1} \\ \vdots \\ w_{b,k} \\ 0 \\ D_i^T N_i^T \varepsilon_{\rho,k} \\ D_i^T N_i^T \varepsilon_{\phi,k} \end{bmatrix} \quad (20)$$

where I and 0 are the identity and zero matrices of appropriate dimensions.

In Equation (20), since W_b (which is a covariance matrix of w_b) is an infinite covariance matrix, the process equation of b_k does not affect the filter solution. Furthermore, the all $a_{i,t}$ for $t \in \{1, \dots, k\}$ are identical. From these two facts, the Equation (20) can be simplified and rearranged as follows:

$$\begin{bmatrix} \overline{D_i^T N_i^T \rho} \\ \overline{D_i^T N_i^T \phi} \end{bmatrix} = \begin{bmatrix} \overline{D_i^T N_i^T B} \\ \overline{D_i^T N_i^T B} \end{bmatrix} \overline{b} + \begin{bmatrix} 0 \\ \lambda I \end{bmatrix} a_i + \begin{bmatrix} \overline{D_i^T N_i^T \varepsilon_\rho} \\ \overline{D_i^T N_i^T \varepsilon_\phi} \end{bmatrix} \quad (21)$$

where $\overline{D_i^T}$ is $blkdiag(D_i^T)$ which is a block diagonal matrix whose all diagonals equal to D_i^T . $\overline{N_i^T}$ and $\overline{B_i^T}$ represent $blkdiag(N_i^T)$, and $blkdiag(B_i^T)$, respectively. $\overline{b}$ is $[b_1^T, \dots, b_k^T]^T$. $\overline{\rho}$ and $\overline{\phi}$ signify $[\rho_1^T, \dots, \rho_k^T]^T$ and $[\phi_1^T, \dots, \phi_k^T]^T$, and their corresponding errors are denoted as $\overline{\varepsilon}_\rho$ and $\overline{\varepsilon}_\phi$.

The fixed baseline solution is identical to the solution which is obtained by solving the Equation (21) under the constraint that a_i is set equal to $\check{a}_i$ (Teunissen, 1999). For a given constraints of $\check{a}_i = z_i$, the fixed baseline solution is as follows:

$$\check{\overline{b}}_i |_{\check{a}_i = z_i} = Q_{\check{b}_i | \check{a}_i} \begin{bmatrix} \overline{D_i^T N_i^T B} \\ \overline{D_i^T N_i^T B} \end{bmatrix}^T Q_{\varepsilon_i}^{-1} \left(\begin{bmatrix} \overline{D_i^T N_i^T \rho} \\ \overline{D_i^T N_i^T \phi} \end{bmatrix} - \begin{bmatrix} 0 \\ \lambda I \end{bmatrix} z_i \right) \quad (22)$$

where $Q_{\varepsilon_i}^{-1}$ is the inverse of covariance matrix of ε_i :

$$Q_{\varepsilon_i}^{-1} = \begin{bmatrix} \beta (\overline{D_i^T \overline{D_i}})^{-1} & 0 \\ 0 & \alpha (\overline{D_i^T \overline{D_i}})^{-1} \end{bmatrix} \quad (23)$$

$Q_{\check{b}_i | \check{a}_i}$ is a covariance matrix of $\check{\overline{b}}_i$ given $\check{a}_i = z_i$:

$$\begin{aligned}
Q_{\check{b}_i|\check{a}_i} &= \left(\begin{bmatrix} \bar{D}_i^T \bar{N}_i^T \bar{B} \\ \bar{D}_i^T \bar{N}_i^T \bar{B} \end{bmatrix}^T Q_{\varepsilon_i}^{-1} \begin{bmatrix} \bar{D}_i^T \bar{N}_i^T \bar{B} \\ \bar{D}_i^T \bar{N}_i^T \bar{B} \end{bmatrix} \right)^{-1} \\
&= \frac{1}{\alpha + \beta} (\bar{B}^T \bar{N}_i \bar{P}_i \bar{N}_i^T \bar{B})^{-1}
\end{aligned} \tag{24}$$

where $\bar{P}_i = \bar{D}_i (\bar{D}_i^T \bar{D}_i)^{-1} \bar{D}_i^T$. Note that by using the definition of $\bar{D}_i$, the $\bar{P}_i$ can also be represented as $I - 1_{n_{\bar{P}_i} \times 1} \left(1_{n_{\bar{P}_i} \times 1}^T \cdot 1_{n_{\bar{P}_i} \times 1} \right)^{-1} 1_{n_{\bar{P}_i} \times 1}^T$ where $n_{\bar{P}_i}$ is the dimension of $\bar{P}_i$. Thus, $\bar{P}_i$ is independent of the pivot satellite, even though $\bar{P}_i$ is defined from $\bar{D}_i$ (Teunissen, 1997).

By substituting Equation (23) and (24) into Equation (22), and by rearranging the equation, the error of fixed baseline solution, under the given constraints of $\check{a}_i = z_i$, can be expressed as follows:

$$\check{\check{b}}_i|_{\check{a}_i=z_i} - \bar{b} = \frac{1}{\alpha + \beta} S_i (\beta \bar{\varepsilon}_\rho + \alpha \bar{\varepsilon}_\phi) + (\bar{B}^T \bar{N}_i \bar{P}_i \bar{N}_i^T \bar{B})^{-1} \bar{B}^T \bar{N}_i \bar{P}_i \lambda (a_i - z_i) \tag{25}$$

where $S_i = (\bar{B}^T \bar{N}_i \bar{P}_i \bar{N}_i^T \bar{B})^{-1} \bar{B}^T \bar{N}_i \bar{P}_i \bar{N}_i^T$. In this equation, the first term, which results from the measurement errors, is a random variate assumed to be normally distributed. The second term, which is induced by the constraints of $\check{a}_i = z_i$, is a bias, and it is zero only when $z_i = a_i$ (i.e., correct fix). Note that, $\check{\check{b}}_i$ is a fault free fixed solution of $\bar{b}$, which comprises the baseline vectors for times 1 through k , under H_i hypothesis. Thus, at time epoch k , the last element of $\check{\check{b}}_i$ is our main focus. It is also important to note that the Equation (24) and (25) show that the conditional $\check{\check{b}}_i$ and its covariance matrix are independent of the choice of pivot satellite.

3.2. Covariance Matrix of the Detection Statistic Given $\check{a}_i$ and $\check{a}_0$

This section shows that, given $\check{a}_i = a_i$ and $\check{a}_0 = a_0$, the covariance matrix of solution separation can be expressed as $Q_{\check{b}_i|\check{a}_i} - Q_{\check{b}_0|\check{a}_0}$. From the Equation (25), the conditional solution separation ($\bar{\Delta}_i$) under the given condition that $\check{a}_i = a_i$ and $\check{a}_0 = a_0$ can be expressed as follows:

$$\begin{aligned}
\bar{\Delta}_i|_{\check{a}_i=a_i, \check{a}_0=a_0} &= \check{\check{b}}_i|_{\check{a}_i=a_i} - \check{\check{b}}_0|_{\check{a}_0=a_0} \\
&= \frac{1}{\alpha + \beta} (S_i - S_0) (\beta \bar{\varepsilon}_\rho + \alpha \bar{\varepsilon}_\phi)
\end{aligned} \tag{26}$$

Its covariance matrix can be expressed as:

$$Q_{\bar{\Delta}_i|\check{a}_i, \check{a}_0} = \frac{1}{\alpha + \beta} (S_0 S_0^T - S_0 S_i^T - S_i S_0^T + S_i S_i^T) \tag{27}$$

This equation can be obtained by using the fact that ε_ρ and ε_ϕ follow the zero-mean normal distribution with the covariance matrices of $\frac{1}{\alpha} I_m$ and $\frac{1}{\beta} I_m$.

The first term in Equation (27) can be written as:

$$\begin{aligned}
S_0 S_0^T &= \left(\bar{B}^T \bar{N}_0 \bar{P}_0 \bar{N}_0^T \bar{B} \right)^{-1} \bar{B}^T \bar{N}_0 \bar{P}_0 \bar{N}_0^T \cdot \bar{N}_0 \bar{P}_0^T \bar{N}_0^T \bar{B} \left(\bar{B}^T \bar{N}_0 \bar{P}_0 \bar{N}_0^T \bar{B} \right)^{-T} \\
&= \left(\bar{B}^T \bar{P}_0 \bar{B} \right)^{-1} \bar{B}^T \bar{P}_0 \cdot \bar{P}_0^T \bar{B} \left(\bar{B}^T \bar{P}_0 \bar{B} \right)^{-T} \\
&= \left(\bar{B}^T \bar{P}_0 \bar{B} \right)^{-1} \bar{B}^T \bar{P}_0 \bar{B} \left(\bar{B}^T \bar{P}_0 \bar{B} \right)^{-T} \\
&= \left(\bar{B}^T \bar{P}_0 \bar{B} \right)^{-1} \\
&= (\alpha + \beta) Q_{\bar{b}_0 | \bar{\alpha}_0}
\end{aligned} \tag{28}$$

The first equality follows from the definition of S_0 , while the second follows from $\bar{N}_0 = I$. The third line follows from $\bar{P}_0 \bar{P}_0^T = \bar{P}_0$ for all i , which can be easily derived from the definition of $\bar{P}_0$. The last equality follows from the Equation (24).

Similarly, the last term in Equation (27) becomes:

$$S_i S_i^T = (\alpha + \beta) Q_{\bar{b}_i | \bar{\alpha}_i} \tag{29}$$

When deriving this equation, the fact that $\bar{N}_i \bar{N}_i^T = I$ is used.

The second term in Equation (27) can be written as:

$$\begin{aligned}
S_0 S_i^T &= \left(\bar{B}^T \bar{P}_0 \bar{B} \right)^{-1} \bar{B}^T \bar{P}_0 \cdot \bar{N}_i \bar{P}_i^T \bar{N}_i^T \bar{B} \left(\bar{B}^T \bar{N}_i \bar{P}_i \bar{N}_i^T \bar{B} \right)^{-T} \\
&= \left(\bar{B}^T \bar{P}_0 \bar{B} \right)^{-1} \bar{B}^T \bar{N}_i \bar{P}_i \bar{N}_i^T \bar{B} \left(\bar{B}^T \bar{N}_i \bar{P}_i \bar{N}_i^T \bar{B} \right)^{-T} \\
&= \left(\bar{B}^T \bar{P}_0 \bar{B} \right)^{-1} \\
&= (\alpha + \beta) Q_{\bar{b}_0 | \bar{\alpha}_0}
\end{aligned} \tag{30}$$

The first equality follows from the definition of S_0 and S_i , while the second follows from $\bar{P}_0 \bar{N}_i \bar{P}_i^T = \bar{N}_i \bar{P}_i$ which will be derived in Appendix. The last equality follows from the Equation (24).

In a similar fashion, the third term in Equation (27) becomes:

$$S_i S_0^T = (\alpha + \beta) Q_{\bar{b}_0 | \bar{\alpha}_0} \tag{31}$$

Finally, by substituting Equations (28), (29), (30), and (31) into Equation (27), we can have the following relation:

$$Q_{\bar{\Delta}_i | \bar{\alpha}_i, \bar{\alpha}_0} = Q_{\bar{b}_i | \bar{\alpha}_i} - Q_{\bar{b}_0 | \bar{\alpha}_0} \tag{32}$$

Since $\bar{\Delta}_i$ is defined by the difference between $\bar{b}_i$ and $\bar{b}_0$, it contains all solution separations for times 1 through k . The Δ_i at current time epoch k is the last element of $\bar{\Delta}_i$. Thus, from Equation (32), $Q_{\Delta_i | \bar{\alpha}_i, \bar{\alpha}_0} = Q_{\bar{b}_i | \bar{\alpha}_i} - Q_{\bar{b}_0 | \bar{\alpha}_0}$ holds valid.

4. VALIDITY OF THE OVERBOUNDING METHOD TO BOUND INTEGRITY RISK

The integrity risk in a navigation system is calculated based on the covariance matrices of the baseline and ambiguity solutions. To ensure reliable evaluations, these covariance matrices, which are derived from the statistical models of measurement errors, must be accurately characterized (Langel et al., 2016). However, the measurement models are usually approximated from the collected data without careful consideration of their tail probabilities. This might make the integrity risk assessment be overly optimistic, potentially leading to hazardous consequences in safety.

In code-based navigation systems, several measurement overbounding methods have been developed to provide an upper bound on the integrity risk when the probability distributions of measurement errors are unknown, but can be bounded (Rife & Gebre-Egziabher, 2007; Langel et al., 2021). The integrity risk in code-based systems is calculated using a one-dimensional integral over a fixed range, where the position error exceeds the protection level. Thus, the primary concern lies in the integrand. However, the integrity risk assessment in a Carrier-phase Differential GNSS (CDGNSS) requires a multi-dimensional integral to compute the probability of incorrect fixes. Moreover, the multi-dimensional integration region is not constant but is a function of the float ambiguity covariance matrix. These complexities in integrity risk evaluation of CDGNSS might invalidate the overbounding method developed for code-based systems.

This section demonstrates that the overbounding method, when combined with the conservative assumption on the integrity risk associated with incorrect ambiguity fixes used for Equation (6), provides the upperbound on the integrity risk. The integrity risk of CDGNSS, in Equation (7), is calculated from the $P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL | \check{a}_i = a_i, H_i)$ and $P(\check{a}_i = a_i | H_i)$. The first probability is derived from the covariance matrix of the conditional fixed baseline solution, while the second is derived from the covariance matrix of the float ambiguity solution. Thus, we need to establish an upper bound for $P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL | \check{a}_i = a_i, H_i)$ and a lower bound for $P(\check{a}_i = a_i | H_i)$ to upper bound the integrity risk. Sections 4.1 and 4.2 demonstrate that the covariance matrices, derived from the overbounded measurement model, provide the upper bound for $P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL | \check{a}_i = a_i, H_i)$ and the lower bound for $P(\check{a}_i = a_i | H_i)$, respectively.

4.1. Upper Bound on $P(|\check{b}_i - b| + T_i > VPL | \check{a}_i = a_i, H_i)$

In code-based system, the integrity risk under H_i hypothesis, when using SS-RAIM, is computed from the following:

$$P(|\hat{b}_{code,i,v} - b_v| + T_{code,i,v} > VPL | H_i) \quad (33)$$

where $\hat{b}_{code,i}$ represents a fault free subset solution against H_i hypothesis, and $T_{code,i}$ is a threshold of its detection statistic. This probability is computed from the covariance matrix of $\hat{b}_{code,i}$ which is derived from the code measurement error model. The overbounding method ensures, even when the exact code measurement error model is unknown, that (Langel et al., 2021):

$$P(|\hat{b}_{code,i,v} - b_v| + T_{code,i,v} > VPL | H_i) < P_{OB}(|\hat{b}_{code,i,v} - b_v| + T_{code,i,v} > VPL | H_i) \quad (34)$$

where $P_{OB}(\cdot)$ is the probability that is computed from the covariance matrix derived from the overbounded code measurement error model.

Let us recall the Equation (21) and (22). Once the $\check{a}_i$ is given as a deterministic value, it is possible to remove the ambiguity present in the carrier phase measurements by subtracting $\check{a}_i$ from them. These ambiguity-corrected carrier phase measurements can then be employed as highly precise pseudorange measurements to compute the fixed baseline solution. In other words, the fixed baseline solution, when $\check{a}_i$ is given as z_i , is identical to the solution obtained by solving Equation (21) with the constraint that a_i is set equal to z_i (Teunissen, 1999). The process to obtain the baseline solution from the ambiguity-corrected carrier phase measurements is exactly the same as the solution computation process in a code-based system. Therefore, the overbounding method, which was developed for the code-based system, is still valid:

$$P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL | \check{a}_i = a_i, H_i) < P_{OB}(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL | \check{a}_i = a_i, H_i) \quad (35)$$

Note that, in Equation (35), $P_{OB}(\cdot)$ is the probability that is computed from the covariance matrix derived from the overbounded code and “carrier phase” measurement error models. This is because the carrier phase measurement as well as code measurement is used to compute $\check{b}_i$.

4.2. Lower Bound on $P(\check{a}_i = a_i | H_i)$

4.2.1. The probability of correct ambiguity fix

In this work, the Integer Bootstrapping (IB) method is used to convert the float ambiguity solution into an integer value. The IB method provides a closed-form expression for the probability of the ambiguity correct fix, $P(\check{a}_i = a_i | H_i)$. This probability is computed by integrating the probability density function (PDF) of float ambiguity solution over the correct fix pull-in region ($B_{Q_{\hat{a}_i}, 0}$) (Teunissen, 2006):

$$B_{Q_{\hat{a}_i}, 0} = \left\{ x \in \mathbb{R}^{n_{a_i}} \mid \left| c_j^T \cdot L_{Q_{\hat{a}_i}}^{-1} \cdot x \right| \leq \frac{1}{2} \text{ for } j = 1, \dots, n_{a_i} \right\} \quad (36)$$

where n_{a_i} is the number of ambiguity elements of $\hat{a}_i$, and is equivalent to $m - n_i - 1$. $L_{\hat{a}_i}$ is a unique unit lower triangular matrix which results from decomposing the covariance matrix of $\hat{a}_i$ (i.e., $Q_{\hat{a}_i} = L_{\hat{a}_i} \cdot D_{\hat{a}_i} \cdot L_{\hat{a}_i}^T$). c_j is a column vector whose elements are all zero but the j^{th} element is one. The $P(\check{a}_i = a_i | H_i)$ can be expressed as follows (Teunissen, 2006):

$$P(\check{a}_i = a_i | H_i) = \int_{B_{Q_{\hat{a}_i}, 0}} f_{\hat{a}_i}(x) dx \quad (37)$$

where $f_{\hat{a}_i}(x)$ is a PDF of $\hat{a}_i$. Assuming the measurement errors follow a normal distribution, $f_{\hat{a}_i}(x)$ becomes the PDF of a zero mean-multivariate normal distribution with the variance matrix of $Q_{\hat{a}_i}$, under H_i hypothesis.

However, in practice, the exact $Q_{\hat{a}_i}$ is unknown since it entirely derives from the statistical model of measurement errors, which is not exactly known. This leads to a situation where $\hat{a}_i$ follows a normal distribution with a true variance matrix of $\Sigma_{\hat{a}_i}$ (but, unknown) and the pull-in region is determined based on the $Q_{\hat{a}_i}$ which is derived from the measurement error models. Therefore, the real $P(\check{a}_i = a_i | H_i)$ can be expressed as (Teunissen, 2006):

$$P(\check{a}_i = a_i | H_i) = \int_{B_{Q_{\hat{a}_i}, 0}} N(0, \Sigma_{\hat{a}_i}) dx \quad (38)$$

where $N(\mu, \Sigma)$ is a PDF of a normal distribution with mean μ and covariance matrix Σ .

Figure 3 shows the difference between theoretical correct fix probability (i.e., Equation (37)) and real correct fix probability (i.e., Equation (38)) in two-dimensional case, taken as an example. Each axis represents each float ambiguity, and the red parallelogram depicts the correct fix pull-in region. The ellipses illustrate the PDF of $\hat{a}_i$. It is important to note that while the integrands of Equation (37) and (38) are different, the integration regions are the same.

In general, the relation between $Q_{\hat{a}_i}$ and $\Sigma_{\hat{a}_i}$ is unknown. However, if the $Q_{\hat{a}_i}$ is derived from the overbounded measurement error models, the prior work associated with the overbounding methods ensures that $Q_{\hat{a}_i}$ is greater than $\Sigma_{\hat{a}_i}$ in a positive-semi definite sense. The following section demonstrates that when the $Q_{\hat{a}_i}$ is greater than $\Sigma_{\hat{a}_i}$ in a positive-semi definite sense, the theoretical correct fix probability is lower than the real one.

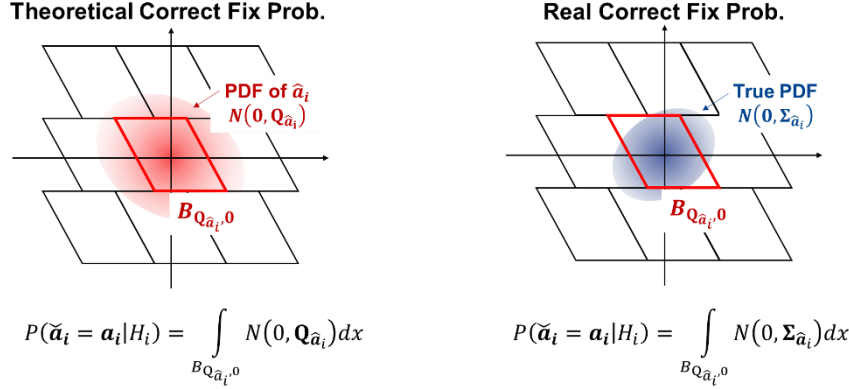


Figure 3. A theoretical correct fix probability and real correct fix probability

4.2.2. Proof for the lower bound of probability of correct fix

From the corollary of Anderson's theory (Anderson, 1955), the following statement holds:

if 1) $\Sigma_{\hat{a}_i} \leq Q_{\hat{a}_i}$ in positive – semi definite sence, and 2) $B_{Q_{\hat{a}_i}, 0}$ is a convex set, symmetric about the origin,

$$\text{then } \int_{B_{Q_{\hat{a}_i}, 0}} N(0, Q_{\hat{a}_i}) dx \leq \int_{B_{Q_{\hat{a}_i}, 0}} N(0, \Sigma_{\hat{a}_i}) dx \quad (39)$$

The first condition stating that $\Sigma_{\hat{a}_i} \leq Q_{\hat{a}_i}$ in a positive-semi definite sense can be satisfied by applying the overbounding method (Langel et al., 2021). Consequently, by demonstrating that $B_{Q_{\hat{a}_i}, 0}$ is a convex set, and symmetric about the origin, we can guarantee that the theoretical $P(\check{a}_i = a_i | H_i)$, which is computed from the covariance matrix derived from the overbounded measurement error model, is a lower bound of the real $P(\check{a}_i = a_i | H_i)$.

Let us recall the definition of $B_{Q_{\hat{a}_i}, 0}$ in Equation (36).

- *Convexity of $B_{Q_{\hat{a}_i}, 0}$:*

Let us pick two arbitrary points $x_1, x_2 \in B_{Q_{\hat{a}_i}, 0}$ and define an interior point $x_3 = \alpha \cdot x_1 + (1 - \alpha) \cdot x_2$ where $\alpha \in [0, 1]$. Then

$$\begin{aligned} |c_j^T \cdot L_{\hat{a}_i}^{-1} \cdot x_3| &= |\alpha \cdot c_j^T \cdot L_{\hat{a}_i}^{-1} \cdot x_1 + (1 - \alpha) \cdot c_j^T \cdot L_{\hat{a}_i}^{-1} \cdot x_2| \\ &\leq \alpha |c_j^T \cdot L_{\hat{a}_i}^{-1} \cdot x_1| + (1 - \alpha) |c_j^T \cdot L_{\hat{a}_i}^{-1} \cdot x_2| \\ &\leq \alpha \cdot \frac{1}{2} + \frac{(1 - \alpha)1}{2} \\ &\leq \frac{1}{2} \end{aligned} \quad (40)$$

The first equality originates from the definition of x_3 . The second inequality follows from the Cauchy-Schwarz inequality, and the third follows from the definition of $x_1, x_2 \in B_{Q_{\hat{a}_i}, 0}$. Therefore, x_3 is in $B_{Q_{\hat{a}_i}, 0}$, which implies that $B_{Q_{\hat{a}_i}, 0}$ is a convex set.

- *Symmetry of $B_{Q_{\hat{a}_i}, 0}$:*

Let us pick an arbitrary point $x \in B_{Q_{\hat{a}_i},0}$. Then,

$$\begin{aligned} |c_j^T \cdot L_{\hat{a}_i}^{-1} \cdot (-x)| &= |c_j^T \cdot L_{\hat{a}_i}^{-1} \cdot x| \\ &\leq \frac{1}{2} \end{aligned} \quad (41)$$

The second inequality follows from the definition of $x \in B_{Q_{\hat{a}_i},0}$. Therefore, $-x$ is in $B_{Q_{\hat{a}_i},0}$, which implies that $B_{Q_{\hat{a}_i},0}$ is symmetric about the origin.

Section 4.1 and 4.2 demonstrated that the covariance matrices, derived from the overbounded measurement model, provide the upper bound for $P(|\check{b}_{i,v} - b_v| + T_{i,v} > VPL | \check{a}_i = a_i, H_i)$ and the lower bound for $P(\check{a}_i = a_i | H_i)$, respectively. Therefore, we can ensure that the measurement model overbounding method provides the upper bound on the integrity risk of CDGNSS system without the exact knowledge of the measurement model.

It is essential to note that since the pull-in regions of the other integers of $z_i \neq a_i$ ($B_{Q_{\hat{a}_i},z_i}$) are convex sets but not symmetric about the origin, we cannot employ the Anderson's theorem. Consequently, we cannot ensure that the theoretical $P(\check{a}_i = z_i)$ is smaller (or, even larger) than the real value. This uncertainty implies that we cannot guarantee the overbounding method provides an upper bound on the integrity risk. Therefore, in CDGNSS system, the overbounding method is valid only when combined with the conservative assumption stating that any event of incorrect ambiguity fixing leads to a loss of integrity.

5. PERFORMANCE EVALUATION

5.1. Simulation Environments

Performance analyses for an illustrative example of dual-constellation, dual-frequency navigation system are carried out to evaluate the developed SS-RAIM based architecture for CD-GNSS system. The simulation parameters are listed in Table 1. In this simulation, the multipath errors and differential ionospheric delay errors are considered for the double differenced measurement error sources. While these errors are usually modeled as first order Gauss-Markov processes to account for their temporal correlations, in this study, they are assumed to have no temporal correlations and to follow the zero-mean normal distributions. For their variances, the overbounded measurement models, which bound the true model used in the simulation, were used. The differential tropospheric delays which are a function of the altitude are assumed as to be negligible, based on the assumption that the difference in altitude between the reference station and the user is small.

Table 1. Simulation parameters

Parameters		Values
Navigation Requirements	Vertical integrity risk (I_{req})	10^{-7}
	Continuity risk under fault free (P_{FA})	10^{-5}
Faults	Occurrence probability	$10^{-5}/SV$
	The number of maximum simultaneous faults	Single
Environments	Location	Daejeon, South Korea
	Constellation	GPS 24 / Galileo 27
	Frequency	L1/L5 for GPS E1/E5a for Galileo
Measurement Error Models	Code multipath error (σ_e)	2 m
	Carrier multipath error (σ_e)	2 cm
	Ionospheric decorrelation error (σ_I)	4 mm/km

5.2. Protection Level and Monitor Performance

To validate the algorithm developed in this work, simulations are conducted under two scenarios: 1) Fault-free scenario and 2) Fault scenario. In the fault-free scenario, there are no satellite faults and only measurement noise is considered. In contrast, under the fault scenario, a satellite is assumed to be faulted with received measurements from this satellite presumed to be biased. Under these two scenarios, the PLs were compared with the positioning errors, and the detectability of the monitor was analyzed. The baseline distance between the reference receiver and the rover was assumed to be 5 km.

5.2.1. Fault free scenario

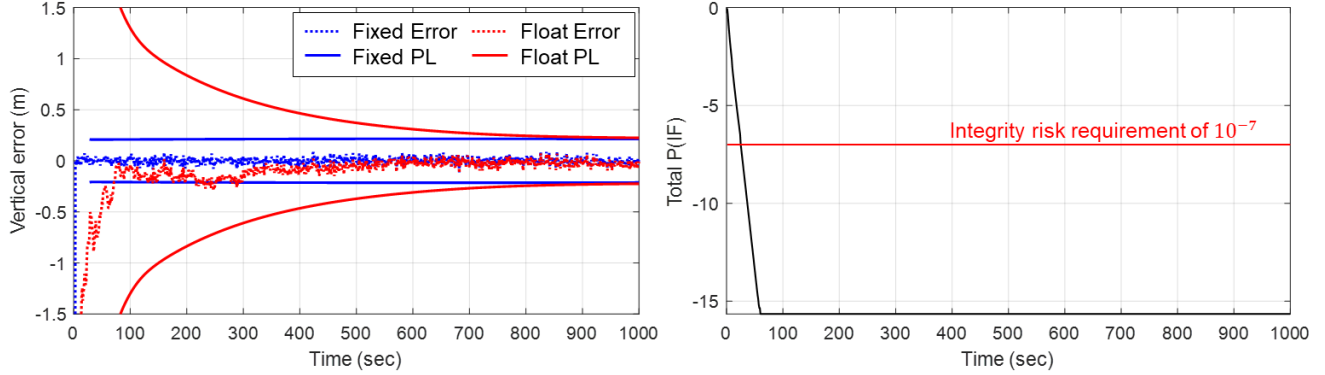


Figure 4. Fault free case simulation results. The left plot shows the vertical estimation error and protection level. The right plot shows the total incorrect fix probability

This section provides a comparison between the fixed solution and the float solution cases to illustrate the benefits of ambiguity resolution. The SS-RAIM algorithm developed in this work was used to compute the protection level for the fixed solution case. For the float solution case, the same navigation filters were employed to compute the float baseline solution, but the conventional SS-RAIM algorithm, developed for code-based systems, was used to compute the protection level. The fixed case is represented by blue lines and the float case by red lines in Figure 4. The dotted and solid lines indicate the vertical baseline errors and their corresponding vertical protection levels, respectively. The results demonstrate that both protection levels bound their corresponding position errors.

In the initial stage of the filtering process, the fixed VPL are significantly smaller than those of the float baseline solution. This reduction is a result of the ambiguity resolution, which significantly decreases the variance of the fixed baseline solution. As discussed in Section 2, fixed VPLs cannot be computed until the total incorrect ambiguity fix probability (as depicted in the right plot of Figure 4) is lower than the specified integrity and continuity risk requirements. Consequently, during the first 28 epochs of the filtering process, fixed VPLs cannot be computed, and thus, the integrity and continuity of the fixed baseline solution cannot be guaranteed. On the other hand, as the float solution does not include the ambiguity resolution process, its VPL can be computed from the onset of the filtering process. Therefore, when the integrity of the fixed position cannot be assured, the float solution can be utilized as a reliable alternative.

5.2.2. Fault scenario

To evaluate the detectability of the SS-RAIM, we simulated a scenario where a satellite is subjected to an ephemeris fault. A step-type bias was injected into the measurements from one assumed-faulted satellite, starting at the 300-second mark and continuing until the end of the simulation. Figure 5 displays the response of the fixed baseline solution and the detection statistics when the inject fault magnitude was 1 meter. The left plot depicts position error (dotted line) and the protection level (solid line), respectively. Points where the monitor successfully identified the faults are marked with red circles. The right plot illustrates the ratios of detection statistics to their corresponding thresholds, with each line denoting each detection statistics. An alert is triggered by the monitor whenever any ratio surpasses 1. The results confirm that the monitor, as designed, can identify faults.

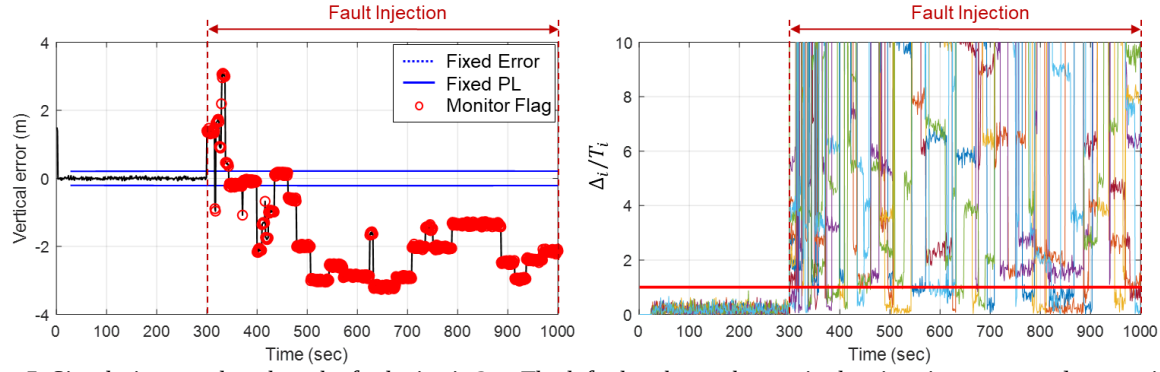


Figure 5. Simulation results when the fault size is 2m. The left plot shows the vertical estimation error and protection level. The right plot shows the ratio of detection statistic and corresponding threshold.

Figure 6 depicts the simulation results when a smaller fault, roughly equivalent to the wavelength of the L1 frequency (0.2 m), was injected. The other simulation parameters remained consistent with those specified for Figure 5. There are instances where the monitor accurately detects the injected faults, whereas at other points the monitor fails to identify these faults. In the instances of monitoring failure, the detection statistics did not exceed the thresholds, and the faulty satellite was still incorporated into the calculation of the fixed baseline solution. These cases, referred to as misdetection, are indicated by blue circles in the left plot of Figure 6. Importantly, even when the monitor failed to detect the faults, the integrity of the system was continuously maintained, as the protection level bounded the position errors of these mis-detection cases.

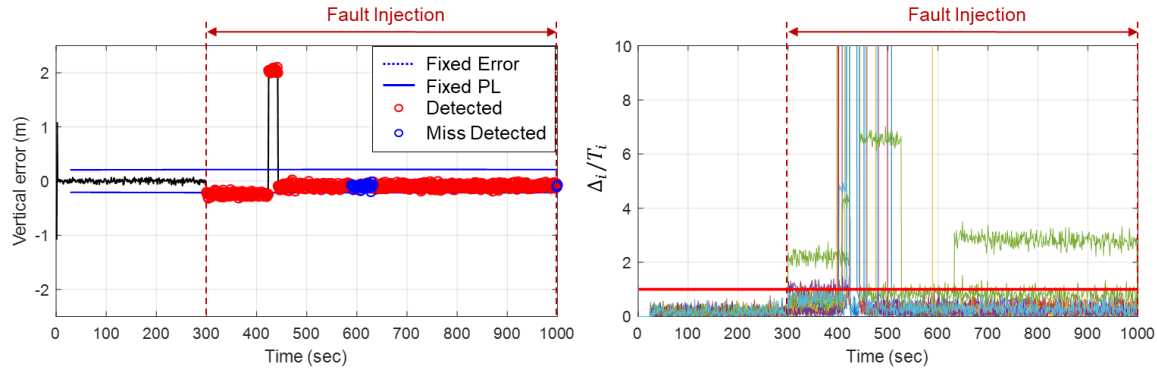


Figure 6. Simulation results when the fault size is 0.2m. The left plot shows the vertical estimation error and protection level. The right plot shows the ratio of detection statistic and corresponding threshold.

6. CONCLUSION

This paper has introduced the SS-RAIM-based integrity architecture for CDGNSS that relies on ambiguity resolution. A solution separation-based test statistic was utilized for fault detection, and the protection level was derived considering the integrity risk associated with the incorrect ambiguity resolution. To apply SS-RAIM, which was originally developed for code-based systems, to CDGNSS, two major concerns were investigated. The concerns arose from differences between the navigation algorithms of CDGNSS and code-based systems. First, the property used to compute the variance of detection statistic for code-based system was verified whether it is directly applicable to CDGNSS systems. This was achieved by deriving the covariance matrix of detection statistics as a function of the covariance matrices of fixed position estimates. Second, it was shown that the overbounding method, when combined with the conservative assumption about the integrity risk associated with incorrect ambiguity fixes, could provide an upper bound on the integrity risk of CDGNSS, even without exact knowledge of measurement error models. Performance analyses for a dual-constellation, dual-frequency navigation system were carried out to evaluate the use of SS-RAIM algorithms in CDGNSS. The protection level of fixed solution was compared with that of float solutions to demonstrate the benefit of ambiguity resolution. The results show that the protection levels of fixed solutions are significantly smaller than those of float solutions. It is achieved through the ambiguity resolution that decreases the position error variances. Additionally, we conducted another simulation to

validate the developed architecture under faulted cases, where bias errors were injected into the measurements of a satellite by assuming a single satellite fault. The simulation results show that while mis-detection cases occur when the fault size is relatively small, position errors are still bounded by the protection level.

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APPENDIX

This appendix shows that $\bar{P}_0 \bar{N}_i \bar{P}_i^T = \bar{N}_i \bar{P}_i$. Since, $\bar{P}_0$, $\bar{P}_i$, and $\bar{N}_i$ are the block diagonal matrices of P_0 , P_i , and N_i , respectively, $\bar{P}_0 \bar{N}_i \bar{P}_i^T$ is $blkdiag(P_0 N_i P_i^T)$ and $\bar{N}_i \bar{P}_i$ is $blkdiag(N_i P_i)$. Therefore, we will show that $P_0 N_i P_i^T = N_i P_i$ to prove $\bar{P}_0 \bar{N}_i \bar{P}_i^T = \bar{N}_i \bar{P}_i$.

From the definition of P_0 , it can be expressed as:

$$\begin{aligned} P_0 &= I - 1_{n_{P_0} \times 1} \left(1_{n_{P_0} \times 1}^T 1_{n_{P_0} \times 1} \right)^{-1} 1_{n_{P_0} \times 1}^T \\ &= I - \frac{1}{n_{P_0}} 1_{n_{P_0} \times n_{P_0}} \end{aligned} \quad (A-1)$$

where n_{P_0} is the dimension of P_0 , $1_{m \times n}$ is a m by n matrix whose elements are all-ones. Note that since I and $1_{n_{P_i} \times n_{P_i}}$ are symmetric matrices, P_i is also a symmetric matrix. By using the Equation (A-1), $P_0 N_i P_i^T$ can be expressed as:

$$\begin{aligned} P_0 N_i P_i^T &= \left(I - \frac{1}{n_{P_0}} 1_{n_{P_0} \times n_{P_0}} \right) N_i P_i \\ &= N_i P_i - \frac{1}{n_{P_0}} 1_{n_{P_0} \times n_{P_0}} N_i P_i \\ &= N_i P_i - \frac{1}{n_{P_i}} \begin{bmatrix} \text{sumrow}(N_i P_i) \\ \vdots \\ \text{sumrow}(N_i P_i) \end{bmatrix} \end{aligned} \quad (A-2)$$

where $\text{sumrow}(A)$ is a row vector which is obtained by summing the elements of matrix A along each row (i.e., the i^{th} element of $\text{sumrow}(A)$ is $\sum_{j=1}^{n_A} A_{ji}$).

From the definition of N_i , $N_i P_i$ can be expressed as:

$$N_i P_i = \begin{bmatrix} 0_{n_i \times n_{P_i}} \\ P_i \end{bmatrix} \quad (A-3)$$

where n_i is the number of faulted satellites and n_{P_i} is the dimension of P_i . Since $P_i = I - \frac{1}{n_{P_i}} \mathbf{1}_{n_{P_i} \times n_{P_i}}$ from the Equation (A-1), we can easily see that $\text{sumrow}(N_i P_i)$ is a zero vector. Therefore, $P_0 N_i P_i^T = N_i P_i$ from the Equation (A-2).

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